

SKILLS

- **Programming Languages:** Python, SQL
- **ML/DL Frameworks:** PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn
- **Tools & Libraries:** Pandas, NumPy, Matplotlib, NLTK, spaCy, OpenCV, SentenceTransformers, LangChain
- **Speech & Vision:** Whisper, Wav2Vec2, PyAnnote, SpeechT5, Tortoise-TTS, Qwen-VL, PaddleOCRVL, trOCR
- **MLOps & Deployment:** MLflow, Docker, vLLM, FastAPI, Celery, Redis, DVC, W&B, ZenML, Gradio, Git
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda), Linux, CI/CD, FAISS, ChromaDB
- **Concepts:** ASR, TTS, OCR, LLMs, RAG, Agents, Speaker Diarization, PEFT/LoRA, Supervised/Unsupervised Learning
- **Languages:** Nepali (Native), English (Professional), Hindi (Proficient)

EXPERIENCE

- **Vertex Special Technology (P.) Ltd.** Kathmandu, Nepal
AI/ML Engineer *Oct 2025 – Present*
 - **OCR Document Extraction (Scainit):** Built an end-to-end document intelligence pipeline that classifies scanned documents (receipts, invoices, bank statements) and extracts structured data using Qwen-VL and PaddleOCR. Implemented markdown extraction from VLM outputs and parsed results into structured JSON. Stored assets on AWS S3; deployed backend via FastAPI, Celery, and Redis on AWS EC2.
 - **AI Backend Infrastructure:** Developed scalable AI backend using FastAPI, Celery, and Redis with document hashing and idempotency logic to detect duplicates and avoid redundant model inference. Managed asset storage on AWS S3 and tracked data and model versions using DVC.
 - **Nepali Handwritten OCR:** Extended Scainit for localized Nepali bank documents. Created synthetic handwritten datasets using *shirorekha* concatenation techniques. Fine-tuned PaddleVL 1.5 with multi-stage training (synthetic → handwritten), PEFT, encoder freezing, and progressive data scheduling on Lambda Labs A100 machines. Researched diffusion-based data augmentation approaches.
 - **Model Inference & Deployment:** Deployed vision-language models using vLLM for high-throughput inference. Managed GPU memory and concurrency for production-grade document processing workloads on AWS EC2 and Lambda Labs A100 instances.
 - **Cross-functional Collaboration:** Worked closely with designers, PMs, and frontend engineers. Mentored junior interns and conducted code reviews to ensure quality and scalability. Worked along with senior engineers too.
- **Ekghanti Services Pvt. Ltd.** Kathmandu, Nepal
AI/ML Engineer *Sep 2023 – Sep 2025*
 - **Voice AI Systems:** Built real-time STT pipelines with Whisper and PyAnnote for speaker diarization. Used denoisers for audio preprocessing and wav2vec2 for emotion classification. Designed diarization-based transcription system for multi-speaker call audio.
 - **End-to-End ML Pipelines:** Designed, developed, and deployed NLP and speech-based systems (ASR, TTS, OCR) using Whisper, SpeechT5, and Transformers for real-time production use cases.
 - **Dataset Creation, Processing & Model Training:** Curated in-house voice datasets with preprocessing, noise filtering, and alignment for ASR, TTS, and emotion classification. Used DVC for dataset and model versioning with metadata tracking over time. Fine-tuned LLMs (LLaMA 3.2, Gemma) using PEFT, LoRA, and quantization; evaluated traditional ML models (SVM, RF) where appropriate.
 - **MLOps & Deployment:** Deployed LLM, ASR, and TTS models via FastAPI with GPU concurrency, request batching, and memory-efficient inference. Integrated MLflow for experiment tracking, Docker for CI/CD, and deployed LLM-based services on AWS EC2 and physical infrastructure using vLLM + LangChain.
 - **Cross-functional Collaboration:** Worked closely with designers, PMs, and backend engineers. Mentored junior interns and conducted code reviews to ensure quality and scalability.

PROJECTS

- **Scainit – OCR Document Extraction Pipeline:** Built an end-to-end document intelligence system: classified scanned documents (receipts, invoices, bank statements) using a custom classifier, extracted structured content via Qwen-VL and PaddleOCR with markdown parsing into JSON. Backend powered by FastAPI, Celery, and Redis with idempotency via document hashing. Assets stored on AWS S3; models trained on AWS EC2 and Lambda Labs A100 machines; data and model versions tracked with DVC.
- **Multi-Speaker Diarization & Transcription:** Built a Whisper + PyAnnote diarization-aware ASR system for call recordings. Integrated multi-session batching and speaker-attributed transcripts. Deployed via FastAPI with GPU semaphore for scalable inference.
- **Context-Aware Conversational Agent:** Built a Nepali LLM-powered chatbot using RAG (LangChain, FAISS) and Ollama. Implemented semantic search with vector DB and multi-turn memory using Letta for persistent conversational context.
- **Speech-to-Text System:** Built Whisper and Wav2Vec2-based real-time ASR system. Deployed with optimized inference on GPU using batching, async processing, and FastAPI.
- **Text-to-Speech System:** Created multi-speaker TTS system using SpeechT5 and Tortoise with MOS of 3.5+. Integrated custom tokenizer via Sentence Transformers.

EDUCATION

- **Nepal College of Information Technology, Pokhara University** Lalitpur, Nepal
Bachelor of Engineering in Software Engineering; GPA: 3.51 *Jan. 2019 – Sept. 2024*
 - **Thesis:** Nepali Sign Language to Nepali Speech using SOTA ASR & TTS models.
 - **Relevant Courses:** Machine Learning, Deep Learning, NLP, Algorithms, Software Engineering, Data Structures

PUBLICATIONS

- **Nepali Sign Language Recognition:** Research on converting visual sign gestures into Nepali speech using CV + TTS pipeline.
- **Synthetic Handwritten Dataset Creation for Nepali (in progress):** Research on generating synthetic handwritten Nepali text datasets using *shirorekha* concatenation – a character-level concatenation technique to construct valid Nepali words – enabling scalable handwritten data generation for low-resource OCR training.
- **ASR/TTS Review Papers (in progress):** Review and benchmarking of recent advances in Nepali ASR & TTS technologies.